



Faculty of Science and Technology
COM113 Computer Architecture & Organization

Time allowed: 3 hours

Section A
Answer Both Questions

Q.1.

- a) Verify De Morgan's Law by using truth table. That is $\overline{A \cdot B} = \overline{A} + \overline{B}$ (8 marks)
- b) Show that $P + (PQ)'$ is a Tautology using a truth table. (7 marks)
- c) Below is a truth table of full adder with A, B and C-IN as inputs and S, C-OUT as outputs as shown. Derive Boolean algebraic expressions for S and C-OUT. (10 marks)
- Sum(S)
 - Carry(C-OUT).

INPUTS			OUTPUT	
A	B	C-IN	C-OUT	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Total 25 marks

Q.2. Construct the following *logic gates* and their related *Truth Table*, and give an example to explain where the *gates can be applied*:

- AND gate (5 marks)
- NAND gate (5 marks)
- OR gate (5 marks)
- EX-OR gate (5 marks)
- NOT gate (5 marks)

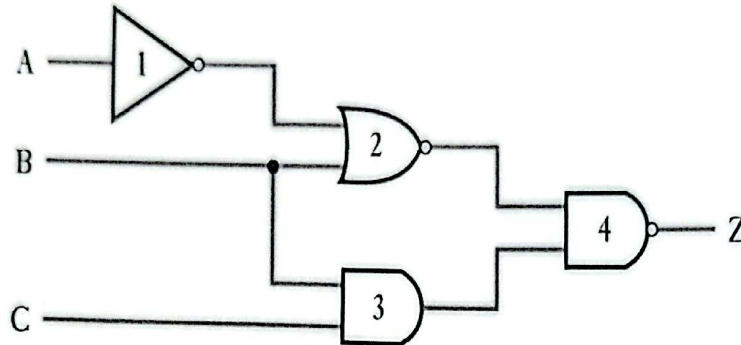
Total 25 marks

Section B

Answer any Two Questions

Q.3. You are provided with circuit diagram given below. Use it to answer the questions that follow:

- Convert the circuit diagram into Boolean equations. (7 marks)
- Represent the simplified expression on a truth table. (8 marks)



- Computer architecture, provides a fundamental understanding on how circuits and hardware communicate within a computer which under stands digital data. List and explain the areas which make use of digital data. (9 marks)

Total 25 Marks

Q.4.

- Convert the following decimal numbers into its binary equivalent. Hence convert the answer to 2's complement. (8 marks)
 - 43_{10}
 - 158_{10}
- Convert the binary the following number into its decimal equivalent. (7 marks)
 - 0110101_2
 - 11010011_2
- Given that $A=1101.0111_2$ Convert the binary number into its. (9 marks)
 - Octal
 - Hexadecimal
 - Decimal

Total 25 Marks

Q.5.

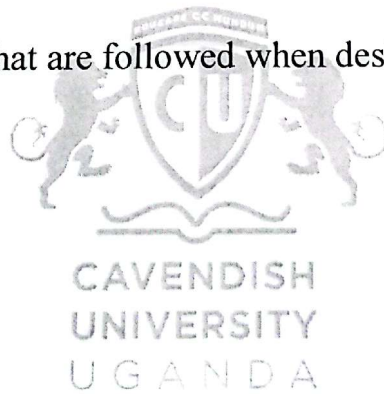
- a) Differentiate between Primary Memory, Secondary Memory and Cache Memory. (9 marks)
- b) Virtual memory has become an important feature in today's computers. Explain the virtual memory concept. (7 marks)
- c) Work out the following without using a calculator
 $11101010_2 + 2122_8 + ABC_{16} - 123_8$, give your answer in decimal. (9 marks)

Total 25 Marks

Q.6.

- a) Explain the following. (8 marks)
 - i. A decoder
 - ii. A ROM
 - iii. An encoder
- b) Give four reasons why it is necessary to study Computer Organization and Architecture. (8 marks)
- c) Mention and explain the steps that are followed when designing a logical diagram. (8 marks)

Total 25 marks





Faculty of Science and Technology
COM112 Computer Architecture and Organization

Time: 9:00am-12:00pm

Time allowed: 3 hours

Section A
Answer Both Questions

Q.1.

- a) Draw the logic diagram for the following: i. NAND gate ii. NOR gate iii. AND gate iv. OR gate **(8marks)**
- b) Discuss the importance of an EX-OR gate. ii. Draw the truth table for EX-OR **(7marks)**
- c) Sketch the block diagram of combinational circuit and discuss the characteristics of combinational circuits **(10marks)**

Total 25 Marks

Q.2.

- a) Write short notes on the following: i. binary number system ii. octal number system iii. Hexadecimal number system **(8marks)**
- b) i. Calculate the Hexadecimal equivalent of 10101_2 ii. calculate the decimal equivalent of $19FDE_{16}$ **(8marks)**
- c) Define a binary code and discuss the categories of binary codes. **(9marks)**

Total 25 Marks

Section B

Answer Any Two Questions

Q.3.

- Microprocessor unit is synonymous to central processing unit (CPU) used in traditional computer. Explain the basic tasks done by a Microprocessor. (7marks)
- Describe the basic architecture of 8085 microprocessor. (9marks)
- Discuss the different instruction formats for the 8085 microprocessor (9marks)

Total 25 Marks

Q.4.

- Memory in a computer system is primarily of two types. Briefly explain these types of memory. (9marks)
- Sketch the memory hierarchy and briefly explain its characteristics. (9marks)
- Explain the fundamental difference between static RAM (SRAM) and dynamic RAM (DRAM) (7marks)

Total 25 Marks

Q.5.

- Explain what is meant by a logic gate and discuss the significance of Logic gates in digital systems. (8marks)
- Draw the truth table for the logic diagram in fig1 below



- Explain the purpose of a truth table in the study of computer architecture and organization.
 - Convert the following switching equation into a truth table:

$$F(A, B, C) = A + BC \quad (9marks)$$

Total 25 Marks

Q.6.

- Sketch the block diagram for a Half adder and explain its characteristics. (8marks)
- Given below is the truth table for a Half adder. Draw the circuit diagram using the truth table.

Inputs		Output	
A	B	S	C
0	0	0	0
0	1	1	0
0	1	1	0
1	1	0	1

(7marks)

- Addressing mode provides different ways for accessing an address to given data to a processor. Briefly explain the Addressing mode for any given processor. (10marks)

Total 25 marks



Faculty of Science & Technology

COM 113 Computer Organisation & Architecture

Date: Wednesday 21st, 2025

Time: 9:00am-12:00pm

Time allowed: 3 hours.

Instructions:

- (i) Read the questions carefully
- (ii) Attempt both sections A and B of the paper as follows;
- (iii) The two numbers in Section A are compulsory and both carry 50% of the total exam marks
- (iv) Attempt any two numbers in section B which carry 50% of the remaining exam marks
- (v) Marks will be awarded for demonstrating proper understanding of methods and answers to questions asked.

**SECTION A
ANSWER BOTH QUESTIONS**

Q.1

In evolution of computers, each generation of computers was designed based on new technological development that would result into better, cheaper, smaller but more efficient machines. With examples at each of the 5 generations of computers, discuss the merits in this statement. **(25 marks)**

Q.2

In a class of Computer Organization and architecture your lecturer makes the following remarks *"Memory hierarchy is about arranging different kinds of storage devices in a computer based on their size, cost and access speed, and the roles they play in application processing. The main purpose is to achieve efficient operations by organizing the memory to reduce access time while speeding up operations"*. With aid of a diagram discuss the computer memory hierarchy including factors mentioned in the passage. **(25 marks)**

SECTION B
ANSWER TWO QUESTIONS

Q.3

- (a) Locality of reference refers to the tendency of the computer program to access the same set of memory locations for a particular time period e.g in loops and subroutine calls. Discuss two different ways this is executed in the Cache memory **(10 marks)**
- (b) Discuss the three cache mapping techniques used to get content from main memory into cache memory. **(15 marks)**

Q.4

With aid of diagrams draw three separate types of cache memory designs and use them to explain how cache performance can be improved. Hint: Use Cache Hit, Cache Miss and Hit Ratio to explain the different design performances **(25 marks)**

Q.5

Addressing modes are an aspect of the instruction set architecture in most central processing unit (CPU) designs. The various addressing modes that are defined in a given instruction set architecture define how the machine language instructions in that architecture identify the operand(s) of each instruction. With aid of diagrams explain how the CPU works with each of the following addressing modes; **(@5marks)**

- (i) Implicit addressing mode
- (ii) Stack addressing mode
- (iii) Immediate addressing mode
- (iv) Direct addressing mode
- (v) Indirect addressing mode

Q.6 Generally, CPU organization is of three types based on the number of address fields. Discuss the three types of CPU organization **(25 marks)**

I did but not sure



Faculty of Science and Technology
COM111 Mathematics For Computing

Date: Thursday August 07, 2025

Time: 10:00am - 1:00pm

Time allowed: 3 hours

Instructions to students:

*Answer **four** questions only*

Answer all questions in Section A and only two questions in Section B.

Write your ID Number

Begin answering each question on a fresh page

There are
answers
in this

Section A

Answer **Both** Questions

Q 1

Kampala Capital City Authority (KCCA) has implemented a smart traffic management system by installing sensors at key intersections—Jinja Road, Kira Road, and Entebbe Road—to collect hourly traffic data. Using this data, a machine learning model has been developed to predict traffic volumes based on the interdependence between roads. The predictions are used not only for traffic flow optimization but also to guide the deployment of infrastructure. High-performance servers are installed at each intersection to process predictions in real time, requiring precise allocations of CPUs, RAM modules, and SSD drives. These hardware requirements are calculated using matrix algebra, ensuring efficient and cost-effective deployment. This system helps KCCA move toward a scalable, data-driven smart city solution. The ML model provides the following system of equations representing interdependent traffic flow:

$$3x_1 + x_2 - 2x_3 = 1$$

$$x_1 + 4x_2 + 3x_3 = 26$$

$$2x_1 - x_2 + 5x_3 = 21$$

Where:

x_1 : Predicted vehicles/hour on Jinja Road

x_2 : Predicted vehicles/hour on Kira Road

x_3 : Predicted vehicles/hour on Entebbe Road

- Write the system of equations in matrix form $AX = B$. (6 marks)
- Solve the system of equations using row reduction to echelon. (10 marks).
- State the application areas of using matrices in real-life situations (9 marks).

Q 2

The curve C has the equation:

$$y = 2x^3 - 9x^2 + 12x - 10$$

- Find the coordinates of the two points on the curve where the gradient is zero. (7 marks)
- The point P lies on curve C, and its x-coordinate is -1. Determine the gradient of C at the point P. Hence, find the equation tangent and normal. (10 marks)
- The point Q lies on curve C such that the gradient at Q is the same as the gradient at point P. Find the coordinates of point Q (8 marks)

$$y = 2x^3 - 9x^2 + 12x - 10$$

$$\frac{dy}{dx} = 6x^2 - 18x + 12$$

$$at x = -1$$

$$\frac{-33}{1} - \frac{1}{36} = \frac{-1186}{36} - 3$$

$$= 24$$

$$y = mx + c$$

$$y = \quad |$$

Section B

Answer Two Questions

Q 3

Let $A = \{3, 5, 7\}$, $B = \{2, 3, 4, 6\}$, and $C = \{2, 3, 4, 5, 6, 7, 8\}$.

- Verify $(A \cap B)' = A' \cup B'$. (6marks)
- Find $(A \cup C) \cap (B \cup C)$. (6 marks)
- Determine $A' \cap (B \cup C)$. (6 marks)
- Use a truth table to show the logical identity:

$$((A \cup B) \cap C)' = (A' \cap B') \cup C'$$
 (7 marks)

Q 4

- Convert the binary number $110101110011.0010101001_{\text{two}}$ to hexadecimal. (8 marks)
- Evaluate: $(3A2 + 1F4) \times 2C$ in hexadecimal. (9 marks)
- Explain the advantages of using base-16 (hexadecimal) in digital computing and memory representation. (8 marks)

Q 5

- Evaluate (8 marks)

$$\int_{\frac{1}{2}}^1 10x\sqrt{1-x^2} dx$$

- Given that: find the value of a (8 marks)

$$\int_0^a (x^2 + 2x - 6) dx = 0$$

- Show clearly that (9marks)

$$\int_3^4 3\sqrt{x} - \frac{4}{\sqrt{x}} dx = k\sqrt{3}$$

Where k is an integer to be found

Q 6

- Convert the number in hexadecimal $2F3.AC_{16}$ to binary. (7 marks)
- Work out the following without using a calculator:

$$11010110_2 + 732_8 + 8E4_{16} - 45_8$$

Give your answer in decimal. (9 marks)

- Explain the advantages of using digital data in the computing world as compared to using analog data. (9 marks)



Faculty of Science & Technology
COM 113 Computer Organisation & Architecture

Instructions:

- (i) Read the questions carefully
- (ii) Attempt both sections A and B of the paper as follows;
- (iii) The two numbers in Section A are compulsory and both carry 50% of the total exam marks
- (iv) Attempt any two numbers in section B which carry 50% of the remaining exam marks
- (v) Marks will be awarded for demonstrating proper understanding of methods and answers to questions asked.

Section A
Answer Both Questions

Q.1

- (i) Without the control Unit in the CPU there is nothing much the computer processor can do. Use 5 (five) reasons, to justify your answers (15 marks)
- (ii) The internal design of the Control Unit of the CPU could be of different formats depending on the intended function. Discuss any two designs highlighting advantages of each design. (10 marks)

Q.2

With aid of a diagram, discuss different memory structure organisation of the computer, explaining the impact of such organisation to computer performance. (25 marks)

Section B
Answer Two Questions

Q.2

✓ The Computer instruction Cycle consists of Fetch, Indirect, Execute and Interrupt cycles. Discuss what exactly happens at each cycle. (25marks)

Q.3

With aid of illustrations explain how different Cache memory designs can be used to improve computer performance. (25 marks)

Q.4

CPU organisation can be inform of Stack, General Register and Accumulator. Discuss step by step of how how each of this organization works (25 marks)

Q.5

With aid of examples, discuss different categories of instructions executed ✓ by the processor (25 marks)



Faculty of Science And Technology
BIT110 Introduction To Information Technology

Date: Friday August 08, 2025
Time allowed: 3 hours

Time: 10:00am - 1:00pm

INSTRUCTIONS TO CANDIDATES

This paper contains two parts: Section A and Section B.

1. **DO NOT** turn examination paper over until instructed to do so
2. Read each question carefully before you answer.
3. Apportion your time according to the marks allocated for the question.
4. Apportion your time equally between Section A and B.
5. Firstly answer the questions that you feel can obtain the most marks for you but, if there are any compulsory questions, you **must** ensure that you attempt these.
6. Number the answers to the questions clearly before answering.
7. Please write as neatly as possible as illegible handwriting cannot be marked.
8. No electronic devices capable of storing and retrieving text may be used.

Section A

Answer Both Questions

Q.1.

- a) What is the difference between data and information? (5 marks)
- b) Explain the characteristics of good information that can be used in an organization. (10 marks)
- c) Examine the factors that determine the choice of a data processing method. (10 marks)

Total 25 Marks

Q.2.

- a) Use your own words to explain what social media means (5 marks)
- b) Compare and contrast WhatsApp and Facebook social media handles in relation to information dissemination. (10marks)
- c) Explain the impact of social media in teaching and learning? (10 marks)

Total 25 Marks

SECTION B
ANSWER TWO QUESTIONS

Q.3.

- a) Define cybersecurity as used in information security and give at least 2 examples of cybercrimes (7marks)
- b) Explain the CIA triad that forms the foundational principles of cyber security(9 marks)
- c) Give at least 3 instances of how people have used social engineering to gain unauthorized access to confidential information. (9 marks)

Total 25 Marks

Q.4.

- a) Identify the different categories of hardware with examples (5 marks)
- b) Suggest reasons why people still use paper-based systems despite the presence of computers. (10 marks)
- c) Explain how internet has facilitated teaching and learning at cavendish (10 marks)

Total 25 Marks

Q.5.

- a) One of the major important qualities of Information is “completeness”. Justify the statement. (6 marks)
- b) Explain any six advantages of off shelf packages over tailor-made software. (10 marks)
- c) Examine the several ways of obtaining IT products and services in an organization. (9 marks)

Total 25 Marks

Q.6.

- a) Differentiate between the following: -
 - i. Low level and high level language (6 marks)
 - ii. Local Area Network and Wide Area Network (6 marks)
 - iii. Full Duplex and half duplex (6 marks)
- b) State the factors you need to consider when selecting an Internet Service Provider (7 marks)

Total 25 Marks

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CAVENDISH UNIVERSITY UGANDA
FACULTY OF SCIENCE AND TECHNOLOGY
DISCRETE MATHEMATICS-DL EXAM

BIT 111

END OF SEMESTER EXAMINATION AUGUST-DECEMBER

Instructions

Read the instructions and understand them

1. You are responsible for managing your time wisely and ensuring that you have enough time to answer all the questions..
2. Plagiarism and cheating are strictly prohibited. Any evidence of academic misconduct will result in immediate disqualification from the exam and further disciplinary action may be taken.
3. Attempt any 4 (Four Questions) of your choice.

Good luck on your exam!

- Simple and Multiple graph

- Isomorphic graphs

What an Euler graph is?

must
closed
trail

use edge only

- Difference b/w Hamiltonian and
Euler graph

Read

Attempt any 4 questions of your choice

Question One:

Each member of a sports club plays at least one of soccer, rugby, or tennis. The following information is known: 43 members play tennis, 11 play tennis and rugby, 7 play tennis and soccer, 6 play soccer and rugby, 84 play rugby or tennis, 68 play soccer or rugby, and 4 play all three sports.

1. Display the information in a Venn diagram. (5 marks)
2. How many members play soccer only? (3 marks)
3. How many members play tennis or soccer, but not rugby? (4 marks)
4. How many members play exactly two sports? (6 marks)
5. How many members play at least one sport? (2 marks)
6. If 60 members play soccer, how many members play tennis? (5 marks)

Total marks: 25

Question Two

How can graph coloring be applied in real-world situations, and what are the challenges and limitations associated with using graph coloring algorithms? Provide specific examples to support your answer. (25 marks)

Question Three

Consider the following propositional logic formula:

$$(A \wedge B) \rightarrow (C \vee \neg D)$$

Where A, B, C, and D are propositions. Assume that A and B are true and C is false.

- a) Create the truth table for the formula. (5 marks)
- b) Using the truth table, determine whether the formula is true or false. (5 marks)
- c) Draw a logic gate diagram for the formula using AND, OR, and NOT gates. (5 marks)
- d) Convert the formula to conjunctive normal form (CNF) and draw the corresponding graph using nodes and edges. (5 marks)
- e) Explain how the graph representation of the formula can be used to find the satisfying assignments for the formula. (5 marks)

Total marks: 25

Question Four

Consider a tree T with n vertices, where n is an odd integer greater than or equal to 3. Let v be a vertex of T . Prove that there exists a vertex u in T such that the distance between u and v is at most $(n-1)/2$.

25 marks:

Question Five

Consider the following symbolic logic statement: $\neg (\exists x)(P(x) \wedge Q(x)) \wedge (\forall y)(R(y) \rightarrow P(y))$

- Translate the statement into English using proper syntax and semantics. (10 marks)
- Identify and explain any potential ambiguities in the translated statement. (10 marks)
- Rewrite the statement in a way that eliminates any ambiguities and maintains the original meaning. (5 marks)

Total marks: 25

Question Six (25 Marks)

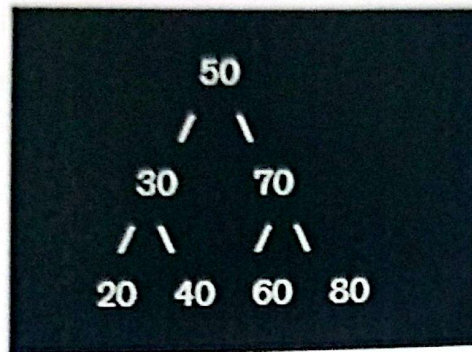
Consider the following propositional formulas:

- $A \wedge B$
- $B \vee C$
- $\neg C \vee D$
- $D \Leftrightarrow E$
- $E \rightarrow A$

Construct a truth table for these formulas and determine the truth value of the compound proposition $((A \wedge B) \vee (\neg C \vee D)) \Leftrightarrow (D \Leftrightarrow E)$.

Question Seven (25 Marks)

Consider the following binary search tree:



- Perform a post-order traversal of the binary search tree and write down the sequence of visited nodes.
- Find the height of the binary search tree.
- Perform a deletion of node 30 from the binary search tree and show the resulting tree.

is a systematic evaluation and analysis of information and ideas to form a decision

Q.1.

- Define critical thinking. (2 marks)
- Imagine you were a campaign manager for one of the Presidential candidates in the Ugandan race this season. You are aware that mass gatherings spark the spread of the corona virus. Design a campaign strategy that creates awareness of the danger of crowds through your manifesto. But address real issues affecting Ugandans at the grassroots. Relate problem solving to critical thinking using the passage as a background. (8 marks)
- The COVID19 pandemic has escalated massive changes in global economies. Lives have changed worldwide. Livelihoods have had to transform. Explain the conditions that must be achieved for exploring problems and creating solutions to those problems in light of recent events following the outbreak of the corona virus. (7 marks)

Total 15 Marks

Q.2.

- Define change by design. (2 marks)
- Draw a divide between critical thinking and innovation as well as how potential employees are agents of change in the 21st century. Design a thinking process that suits the current work environment where there is minimal human contact, and constant use of virtual/digital spaces to relay information. (8 marks)
- Explain the conditions that create a relationship between research and innovation as the most sought after soft skills in the fast paced global economy. Many of which we have recently discovered following new challenges posed by COVID19. (7 marks)

Total 15

Growth model → is the belief that abilities and intelligence can be developed through dedication, effort and learning.

fixed model → is the belief that

Section B
Answer Both Questions

Q.3.

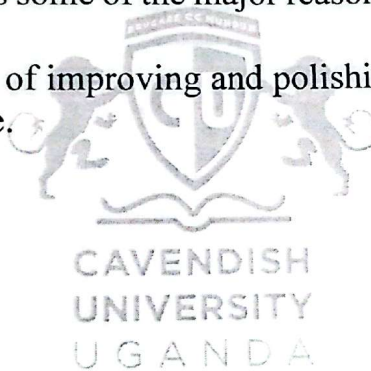
- a) What do you understand by professional body language? (2 marks)
- b) Employers normally have a specific requirement for all target employees within their organizations. What skills do employers look for? (7 marks)
- c) How do you make yourself highly employable, to an extent that your employer can never do without you? (8 marks)
- d) How do you ensure you continue to be an asset to your boss or organization/ company/industry and business entity? (8 marks)

Total 15 Marks

Q. 4.

- a) Define the mindset with examples. (2 marks)
- b) Giving relevant examples, discuss some of the major reasons for the growth mindset in different contexts. (8 marks)
- c) State and explain the major ways of improving and polishing the growth mindset skill particularly for academic excellence. (7 marks)

Total 15 Marks



mindset is an established set of attitudes of a person concerning culture values and

eg - around us during - feeling inspired by other people's success

Section B

Answer Two Questions

Q.3.

- a) How does the growth mindset intersect with other workplace skills, such as agility, creativity, and collaboration? **(10 marks)**
- b) How can these skills complement each other to create a workforce that is prepared for the challenges of the 21st-century economy? **(15 marks)**

Total 25 marks

Q.4.

- a) Define communication styles. **(5 marks)**
- b) Discuss with examples, the 4 types of communication styles. **(10 marks)**
- c) How can organizations encourage cross-cultural collaboration while maintaining a proper communication in a global economy where employees are often spread across different time zones and cultures? Suggest 10 strategies for effective communication skills in the work environment. **(10 marks)**

Total 25 Marks

Q.5.

Define and explain, with examples, the following five elements of the communication process:

- a) The message (5 marks) *eg a text message say "Meet me at 5pm" is the message being conveyed*
- b) The channel (5 marks)
- c) Decoding (5 marks)
- d) Context (5 marks)
- e) Feedback (5 marks)

Total 25 Marks

Q.6

As much as reading, as a process and experience, can be unpleasant, involving and tiresome, it is inevitable and, more importantly, enriching.

- a) Give any five benefits of reading. **(10 Marks)**
- b) Identify and explain all the nine steps of developing the habit of reading for knowledge and skills **(10 Marks)**

Total 25 Marks

with the right skills, mindset, and environment, ordinary students can achieve extraordinary outcomes.

Q.1

a) Reflecting on the transformation of Amina in the story above, explain how the key themes taught in the "Communication Skills and Learning Skills for Employability" course contributed to her personal, academic, and professional development. In your answer, refer to at least five specific course concepts or practices (e.g., communication styles, learning how to learn, critical thinking, professional writing, or digital tools).
(Your essay should be at least 500 words, with clear introduction, body, and conclusion.)
(30 Marks)

SECTION B

ALL QUESTIONS MUST BE ANSWERED

Q.2.

- a) Define effective communication. (5 marks)
b) Use a valid example in each case to illustrate ten ways technology, AI, and digital literacy improve effective communication. (10 marks)

Total 15 Marks

Q.3.

- a) Define perspectives in communication. (5 marks)
b) Using your knowledge of critical thinking and mindset, discuss 10 factors influencing perspectives in communication. Suggest for each factor a mitigating strategy to minimize challenges in future communication endeavors.

(10 marks)

Total 15 Marks

- communication styles
- learning how to learn
- critical thinking
- professional writing
- Digital tools



Faculty Of Socio-Economic Sciences
BJC/DJC/ADHM110 Communication Skills & Learning Skills For
Employability

Date: Saturday August 10, 2025

Time: 10:00am -1:00pm

Time allowed: 3 hours.

INSTRUCTIONS TO CANDIDATES

Answer ALL Questions

CASE STUDY BASED

A Story That Speaks Volumes (Inspired by the Course Themes)

In the bustling corridors of a university in Uganda, a first-year student named Amina nervously walked into her "Communication Skills & Learning Skills for Employability" class. She had dreams of one day working in cyber security, but she lacked confidence in speaking, writing, and interacting in a professional setting.

In Week 1, Amina learned that communication is not just about speaking English well—it's about listening, being clear, understanding others, and expressing herself authentically. She realized communication was a skill she could learn, not a gift you're born with.

By Week 3, after learning about body language, active listening, and how tone influences meaning, Amina began speaking up in class. She started writing clearer emails and participating in group discussions.

In Week 7, the class focused on **growth mindset** and **learning how to learn**. Amina was blown away. She discovered the **Pomodoro technique**, set academic goals for herself, and began journaling her learning journey. She had always feared failure, but now she saw mistakes as stepping stones.

By Week 10, the focus was on **critical thinking**. Using the 5Ws and 1H method (What, When, Where, Who, Why, and How), she began questioning ideas, researching deeply, and contributing to problem-solving exercises in class.

When mock job interviews were conducted in Week 12, Amina impressed the panel with her confident body language, clear responses, and strong eye contact. She was even chosen to represent her cohort in a university innovation competition, showcasing her Excel and PowerPoint skills learned in class.

By the end of the semester, Amina wasn't just more employable—she had become a critical thinker, an active learner, and a more confident communicator. Her transformation proved that

Cavendish University Uganda - Examination, 2025

Page 1 of 2

Section A
Answer Both Questions

Q.1

- a) As students from Mbarara High School raised concerns of bullying in schools, with examples explain how cyber bullying affect the academic Performance of children in schools.(10 marks)
- b) In regards to Patricia Arinaitwe concern, How best can you help these pregnant girls to continue their education at home? (5 marks)
- c) As a student of ICT, how can you use information technology address issues in the society? (10 marks)

Total 25 Marks

Q.2

- d) Mbale Secondary School and Busia Secondary School emphasized that many students in rural communities lack access to technology. What are the likely causes of this challenge? (10 marks)
- e) Much as internet expose children to dangers such as cyber bullying, it has greatly seen as a tool that influence learning in schools. Justify the statement using your experience at cavendish. (15 marks)

Total 25 Marks

Section B
Answer Two Questions

Q.3.

- a) Identify the application software you always use at campus (5 marks)
- b) Explain the competing network topologies that are used to configure networks in an organization. (8 marks)
- c) What are the challenges of tailor-made software in an organization (12 marks)

Total 25 Marks

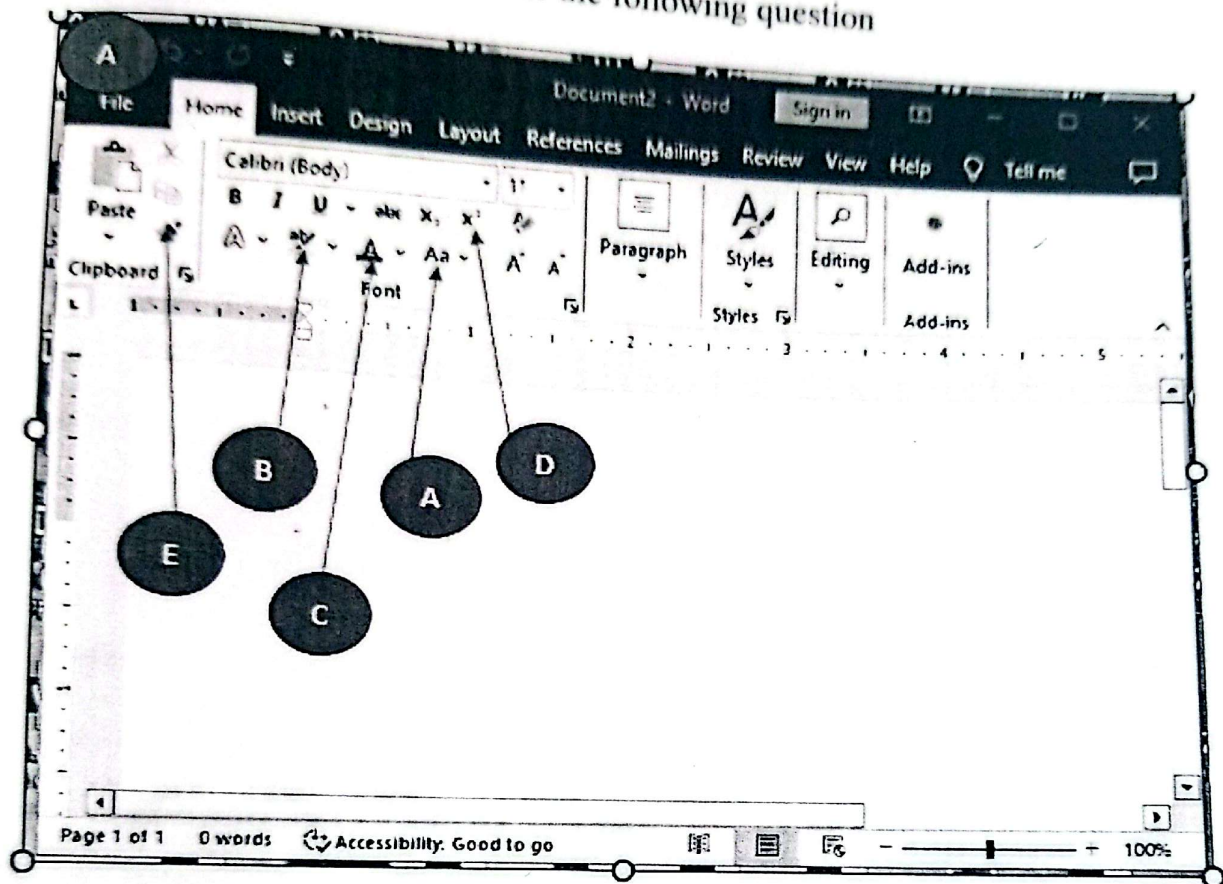
Q.4.

- a) Surveillance has minimized privacy of different people. Justify this statement with examples of incidences that have so far happened. (5 marks)
- b) Identify and explain any 5 cybercrimes that are common today (10 marks)
- c) What would you consider when choosing a smart device to be used in the learning process? (10 marks)

Total 25 Marks

Q.5

Use the diagram below to answer the following question



- a) What is the difference between B and C? (5 marks)
- b) When do you use the following? (4 marks)
 - i. D?
 - ii. E
- c) What are categories of A (5 marks)
- d) Write the steps of achieving the following:-
 - i. Ghost text such as "confidential", "your name" behind the content of the page (3 marks)
 - ii. A border around the page (3 marks)

Total 25 Marks



Faculty of Science and Technology
BIT110 Introduction to ICT

Date: Sunday, December 17th, 2023

Time: 9:00am -12:00Pm.

Section A

Q.1.

- a) In today's technological arena, traditional "Chalk and Talk" teaching pedagogy is used in integration with ICT enabled tools". Discuss this statement with examples. **(10 marks)**
- b) For effective communication with students, University/Institutes uses the various communication tools. Identify those tools that you normally use at work places and explain how they are used during teaching and learning **(20 marks)**
- c) Why do you think corporate organizations refrain from using off the shelf software **(20 marks)**

Total 50 Marks

Section B
Answer Both Questions

Q.2.

- a) How is cost considered as one of the major challenges of using networks in an organization? **(5 marks)**
- b) In today's world of technology, continents have become a global village. Justify **(10 marks)**
- a) WhatsApp is one of the fastest technologies used to exchange digital messages. Assess the challenges of using WhatsApp over other methods of communication. **(10 marks)**

Total 25 Marks

Q.3.

- a) What are the possible reasons why an organization invest in information security? **(10 marks)**
- b) Logical computer security guarantees maximum security to organization's information. Discuss **(5 marks)**
- c) Examine different logical measures that you can employ to ensure information security **(10 marks)**

Total 25 Marks

Q.1

- a) Explain how ICT improved education management and delivery at Bright Future High School? (10 marks)
- b) What factors would Bright Future High School consider to choose computers to be used in the school's ICT project. (10 marks)
- c) Give two examples of keyboard shortcuts that were introduced to improve staff productivity. (5 mark)

Total 25 Marks

Q.2

- a) With examples, differentiate between system software and application software as used in the school's computers. (7 marks)
- b) Give two ways how each of these applications like Microsoft Word, Microsoft Excel and Powerpoint were used in its daily administrative activities? (8 marks)
- c) What security threats are likely to attack Bright Future High School whilst using these file-based systems in 2(a)? (10 marks)

Total 25 Marks



Faculty of Science and Technology
BIT 111 Discrete Mathematics

Date: Monday August 04, 2025

Time: 10:00am - 1:00pm

Time allowed: 3 hours.

INSTRUCTIONS TO CANDIDATES

This paper contains two sections A and B.

Section A is compulsory and carries 50 marks. Attempt two questions from section B
A total of FOUR questions should be attempted.

Read the following before answering the examination questions.

1. **Section A is compulsory**
2. Read each question carefully before you answer.
3. Apportion your time according to the marks allocated for the question.
4. Apportion your time equally between Sections A and B.
5. Firstly answer the questions that you feel can obtain the most marks for you but, if there are any compulsory questions, you **must** ensure that you attempt these.
6. Number the answers to the questions clearly before answering.
7. Please write as neatly as possible as illegible handwriting cannot be marked.
8. No electronic devices capable of storing and retrieving information may be used
9. You are encouraged to make use of the rates of tax at the end of the question paper.

Section A

Answer Both Questions

Q.1 In a college library, four different business newspapers – Economic Times, Business Standard, Business Line and Financial Express – are available. All students visit the library regularly but 20% of them do not read any business newspaper. The four newspapers given in the above order are read by 230, 180, 180 and 220 students respectively. The number of students reading exactly 2 newspapers for any two newspapers is 20. There are 30 students who read all the four newspapers but there is nobody who reads exactly three out of four newspapers.

- a) Draw a venn diagram to represent the distribution of readers for different newspapers (10 marks)
- b) How many students do not read any newspaper at all (5 marks)
- c) What percentage of the people reading Business Standard also read at least one other newspaper? (5 marks)
- d) If all the students in the college including those who do not read any newspaper read at least one newspaper, (Out of the four newspapers above) which he is not reading at present, then what is the least number of students reading all the four newspapers? (5 marks)

Q.2

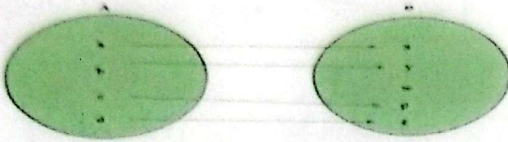
- a) Translate the following English sentences to Propositional Logic. Propositions: (R)aining, Liron is (S)ick, Liron is (H)ungry, Liron is (HA)ppy, Liron owns a (C)at, Liron owns a (D)og
 - i) It is raining if and only if Liron is sick (3 marks)
 - ii) If Liron is sick then it is raining, and vice versa (3 marks)
 - iii) It is raining is equivalent to Liron is sick (3 marks)
 - iv) Liron is hungry but happy (3 marks)
 - v) Liron either owns a cat or a dog (3 marks)
- b) Use a truth table to prove whether $(P + Q)P \rightarrow (P Q + PP)$ (10 marks)

Section B
Answer Two Questions

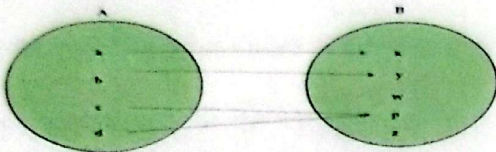
Q.3

(a) Identify the type of functions illustrated in the diagrams here below with justifications as to why you identify them as such **(@5 marks)**

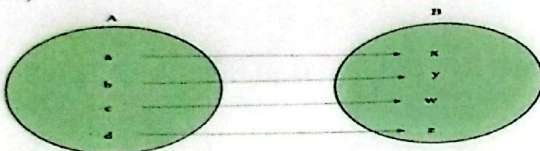
(i)



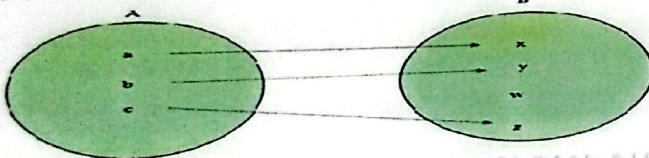
(ii)



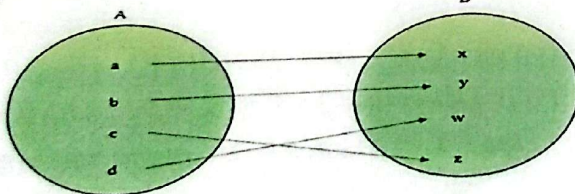
(iii)



(iv)



(v)



(b) In a class of 5 students the result score in Discrete maths and Computer Organization was as follows;

Mark	John	Richard	Luke	Patrick	
80	75	77	85	90	Discrete Maths
86	79	79	82	93	Computer Organisation

(i) Draw a venn diagram representing the relationship between score in Discrete math and Computer Organisation as shown above **(5 marks)**

(ii) Could you refer to the relationship above to be functional in nature? If so what type of functional relationship and Why? **(5 marks)**

Q.4

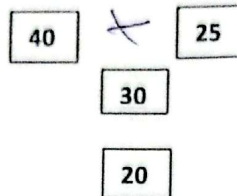
(a) Let R be the relation on $\mathbb{N}$ defined by $x + 5y = 20$, i.e. $R = \{(x,y) : x + 5y = 20\}$.

(i) Write R as a set of ordered pairs

(5 marks)

(ii) Find the inverse of R (R^{-1})

(5 marks)



(b) Determine whether the following relation is an equivalent relation by giving your reason.

$\{(a,b) \mid b \text{ is a supervisor of employee } a\}$ on the set of all employees in Cavendish University (5 marks)

(c) List the ordered pairs in the relation R from $A = \{0,1,2,3,4\}$ to $B = \{0,1,2,3\}$, where $(a,b) \in R$ if and only if

(i) $a + b = 2$ (5 marks)

(ii) $a^2 = 1/2b$ (5 marks)

Q.5.

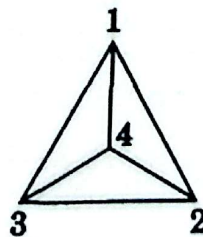
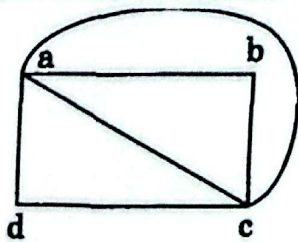
a) Use a truth table to prove whether $\overline{X+Y} \rightarrow \overline{Y} : X \neq X+Y$ (10 marks)

b) Evaluate as to whether $(P+Q) \rightarrow \neg P$ is equivalent to $(P \rightarrow Q) \cdot (Q \rightarrow P)$ (10 marks)

c) Evaluate the truth of the statement "If it is TRUE that $2+5 \neq 8$ or Kampala is in Uganda but not both then London is neither in the UK nor Lagos is in Nigeria but both. (5 marks)

Q.6

a) Determine whether the following graphs are Isomorphic, if so why? (10 marks)

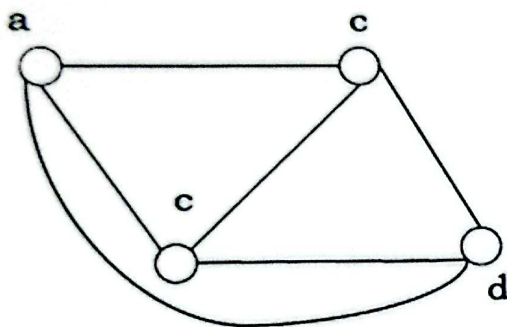


(b) With aid of diagrams explain each of the following graphs @ 2 marks

(i) Complete graph

- (ii) a multi graph
- (iii) Euler graph
- (iv) Null graph
- (v) Simple graph

(c) Draw an adjacency matrix and adjacency list for the following Graph (5 marks)



4

5

10/1/15 (02/11/15) PA 12

$v = \{0\} \rightarrow 20$

(5,1)

$x = 15$

$v = \{1\} \rightarrow 20$

(10,2)

$x = 10$

$v = \{2\} \rightarrow 20$

(15,3)

$x = 5$

UNIVERSITY

UGANDA

$v = \{3\} \rightarrow 20$

(0,4) - 1 Hal of 100

$x = 0$